

Brahma Ganesh Katrapalli

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SUMMARY

Entry-Level ASIC Design Verification Engineer with hands-on experience in SystemVerilog and UVM. Built end-to-end verification environments for AXI4 and APB protocols using constrained-random stimulus, SVA, functional coverage, and scoreboard-based checking. Strong understanding of digital design fundamentals, protocol timing, and debug using waveform analysis.

TECHNICAL SKILLS

HDL & Verification: Verilog, SystemVerilog, UVM (1.2), SVA, Functional Coverage, Constrained Randomization Verification

Digital Design: FSMs, Timing Analysis, Setup/Hold Checks, Synchronization

Protocols (Hands-on): AXI4, APB, UART

Simulation Tools: Synopsys VCS, Verdi, GTKWave

Environment: Linux, Git

Programming: Python (Basics), Shell Scripting (Basics)

PROJECTS

AXI4 Protocol Verification using UVM | SystemVerilog · UVM · SVA · VCS · Verdi *Jan – Feb 2026*

- Verified a custom AXI4 Slave RTL supporting FIXED, INCR, and WRAP burst transactions.
- Built a UVM verification environment comprising agent, sequencer, driver, monitor, scoreboard, and coverage components.
- Generated directed and constrained-random scenarios to validate protocol functionality.
- Implemented SVA checks for VALID/READY handshake, burst transfer, and address alignment conditions.
- Performed regression testing covering reset and error scenarios.
- Achieved 100% functional coverage with all tests passing successfully

APB Slave RTL Design and UVM Verification | SystemVerilog · UVM · SVA · VCS *oct – Nov 2025*

- Evaluated a 256×32 APB Slave RTL supporting read/write transactions, wait states, and PSLVERR generation.
- Created a reusable UVM testbench architecture for protocol verification.
- Assessed SETUP and ACCESS phase timing, handshake behavior, wait-state handling, and error response scenarios.
- Executed directed and constrained-random test cases to exercise corner conditions.
- Applied assertions and functional coverage to ensure protocol compliance.
- Reached 100% functional coverage with all test cases passing.

UART Protocol – UVM Verification | SystemVerilog · UVM · SVA · VCS · Verdi *Feb – March 2026*

- Developed a UVM-based verification environment for full-duplex UART (TX/RX loopback) communication.
- Designed constrained-random and directed test scenarios to validate error conditions including parity and frame failures.
- Implemented scoreboard-based self-checking and SystemVerilog Assertions for protocol validation.
- Achieved full functional coverage closure with regression testing across multiple operating modes.

TRAINING and Project Experience

ASIC Design Verification Training | ChipEdge Technologies *Jun 2025 – Nov 2025*

- Completed training in Verilog, SystemVerilog, UVM, and assertion-based verification; built reusable UVM agents, scoreboards, and coverage models from scratch.
- Executed regression runs, performed coverage analysis, and debugged designs using Synopsys VCS and Verdi in a Linux environment.

EDUCATION

B.Tech in Electronics & Communication Engineering | Bharath University, Chennai *Jun 2021 – May 2025*

CGPA: 8.24 / 10

CERTIFICATION

ASIC Design Verification | ChipEdge Technologies

2025